



Web-Based Street Vendor Finder: Real-Time Tracking and Reviews for Penang Foodies and Tourists

BSc (Hons) Software Engineering 3+0 in collaboration with University of Salford Manchester, UK

Wong Jia Zhun @00752991 SCO6230029BSE



University of
Salford
MANCHESTER

SENTRAL
COLLEGE PENANG

Introduction

This project aims to develop a web-based platform designed to help users discover street vendors in Penang, catering to both local food lovers and tourists. The platform will bridge the gap between street vendors and food enthusiasts by offering a user-friendly interface with features that make it easy to explore the city's street food culture. It will serve as a convenient tool to promote local cuisine and support small businesses.

Aim

The platform will be built with features that include vendor categorization, advanced search and filtering options, real-time location updates, review functionality, and integration with social media. These tools will create a digital guide to Penang's Street food, offering personalized recommendations and encouraging user engagement. By combining these features, the platform will provide a comprehensive experience for foodies and vendors alike.

Motivation

The motivation behind this project is to enhance the visibility of street vendors, especially smaller ones that are harder to locate. By making them more accessible, the platform promotes the vibrant and diverse street food culture of Penang, benefiting both the vendors and those who seek out authentic food experiences. It aims to create a stronger connection between vendors and customers, improving engagement for both sides.

Background

The platform will focus on developing features such as vendor categorization by cuisine, location, and ratings, alongside real-time location tracking. Users can find nearby vendors with a single click and access directions through Google Map integration. Additionally, the platform will provide multilingual support, ensuring it is easy to use for both tourists and locals alike.

Customer

The platform is designed for local food lovers and tourists visiting Penang. Street vendors, especially small or mobile ones, will benefit from increased visibility and ease of access to customers. Tourists and locals will have an easier way to discover hidden food gems and explore Penang's Street food scene in a more organized manner.

Objectives

Main 1: To Develop an Intuitive Web-Based Platform
Main 2: To Implement Location-Based Vendor Discovery
Main 3: To Enable Vendor Location Update Feature
Main 4: To Integrate Google Map API
Main 5: To Provide Multilingual Support

Main 6: To Develop Advanced Search and Filter Options
Main 7: To Create Vendor Profiles and Notifications
Main 8: To Integrate Social Media Sharing
Optional 1: To Develop Real-Time Location Tracking
Optional 2: To Implement User Reviews and Personalized Recommendations

Proposed Methodology

The project will follow an Agile methodology, focusing on iterative development and regular self-assessment to ensure the platform meets the needs of both users and vendors. The process will start with requirement gathering, where data will be collected through surveys and interviews with street vendors, food enthusiasts, and tourists. This approach allows for the prioritization of key features based on real user feedback, with ongoing refinement of these requirements throughout the development process to maintain flexibility.

In the design and development phase, the platform will be built incrementally using JavaScript, PHP, HTML, CSS, and MySQL, with Google Map API integration for location tracking. PhpStorm will be used as the primary development tool. The development will focus first on core features such as location-based vendor discovery, vendor location updates, and multilingual support, with each sprint introducing new functionality or refining existing ones based on feedback.

Testing will be an ongoing part of the project, including unit tests, system tests, and user acceptance tests (UAT) to ensure each feature meets the specified requirements. Once the core functionality is validated, the platform will be deployed. Post-deployment, performance will be monitored, and user feedback will be collected to inform future updates and feature enhancements.

Development Requirements

The web-based platform will be developed on a Windows 10 operating system using PhpStorm (v2024.2.3) as the primary integrated development environment (IDE). PhpStorm is chosen for its robust support for PHP and JavaScript, facilitating an efficient development process for both front-end and back-end components. The core technologies employed include PHP (v8.3) for server-side scripting, JavaScript for client-side functionality, HTML5 for structuring content, and CSS3 for styling the web pages. For database management, MySQL (v8.0) will be utilized to manage user data and vendor information effectively.

For testing and deployment, various web browsers will be used, including Google Chrome, Firefox, and Microsoft Edge. These browsers will help ensure cross-browser compatibility and a seamless user experience across different platforms. Additionally, development will involve testing devices such as computers and smartphones to verify functionality and responsiveness.

In terms of equipment, I will use a workstation with an Intel i5 processor, 48GB RAM, and a dedicated graphics card to support the development environment and handle resource-intensive tasks. Access to computer labs will provide a suitable space for development and testing. Furthermore, reliable internet access will be essential for research, collaboration, and hosting the web application on servers during the development phase. This combination of software, hardware, and facilities will enable efficient development and testing of the platform.

